

## Assignment 4

1. Show that  $Z_4 \oplus Z_6 \simeq Z_2 \oplus Z_{12}$
2. Show that if  $G$  is a finite group and  $H$  is a proper subgroup of  $G$ , then  $G$  cannot be written as a union of conjugates of  $H$ .
3. Show that any finite group with more than 2 elements has a non-trivial automorphism.
4. Prove that any finite group of even order contains an element of order 2.  
Hint: Show that  $t(G) = \{g \in G : g^2 \neq e\}$  has an even number of elements.
5. Show that if  $G/Z(G)$  is cyclic, then  $G$  is abelian.